1. Process Flow

1.1. Sample Arm

1.1.1. Pick New Tip

1.1.1.1. The PANTHER Instrument uses Eppendorf Tips, Part Number 901121.

1.1.1.2. The command sent to pick or strip a tip begins with the first two bytes “FD 7E” or “FD 7F”, 7E if sample arm, 7F if reagent arm. Pick tip commands have the sixth byte as 01 (eject tip is 00h).

1.1.1.3. The pipettor verifies that a tip was picked before it proceeds to the TCR addition step. If no tip is present, it attempts to pick a tip at the next tip position. If it fails to pick a tip three times in a row, it will cancel the reagent addition step and invalidate the remainder of the tip tray (mark it as empty in the GUI).

1.1.2. TCR Addition

1.1.2.1. Liquid level is detected using B-LLD or C-LLD, depending on the assay sequence file. If C-LLD, the capacitive trace for the level detection step will be logged and saved in the C-LLD (See Section 6.1.3) curve file. The detected liquid height is compared to an expected value. If the value is within the appropriate limits as determined by sequence file parameters and previous level detections, the sample arm continues processing. If not within appropriate limits, the sample arm will cancel TCR addition and reschedule the tests intended for that MTU to the next available MTU.

1.1.2.2. TCR is aspirated, volume determined by number of tests scheduled for the current MTU. The pressure observed by the transducer is recorded to “AspirationPressureCurve_...” file.

1.1.2.3. The pipettor retracts to an offset height from the detected liquid surface. There are then a few dispenses, or “Spitbacks”, of TCR back into the reagent bottle. This is done to prime the tip and remove any lead screw backlash from the aspiration movement of the piston. There are no process controls on these dispenses as it is expected that they are not consistent with assay dispenses of TCR. The number of spitbacks is sequence file dependent.

1.1.2.4. The pipettor moves to the MTU and dispenses TCR into each of the scheduled MTU tubes. This is done in a multi-dispense manner with no LLD check afterwards. There are RDV process controls for these dispenses. If a fault is detected during the dispense, the tube where the fault occurred is flagged as “RDFT” and the tests scheduled for this MTU are rescheduled for the next available MTU. Up to 3 RDFT errors can happen with rescheduling, then the operator is requested to remediate the TCR error by loading a new kit or checking the bottle for precipitates or bubbles.

1.1.2.5. If the necessary volume of TCR is greater than that which can fit in the tip (over ~160ul/MTU tube), then the pipettor will make the necessary number of trips to/from the TCR bottle to transfer the required volume. E.g. for the Ultrio Elite assay, 400ul of TCR is needed per MTU tube; the pipettor dispenses 3 aliquots of TCR to/from the TCR bottle using the same tip for this reagent addition step.

1.1.3. Strip Tip / Pick New Tip

1.1.3.1. After successfully dispensing the necessary volume of TCR into each MTU tube, the PANTHER Instrument will strip the tip used for TCR addition.

1.1.3.2. The Sample Pipettor will move to either the front or rear Sample Eject Slot and rapidly descend until the eject mechanism is ~11 mm from the top of the Eject Slot. From this position, it moves at a slower speed (to increase the torque available to the z-motor) the remainder of the 11 mm and ejects the disposable tip where it then falls down the Eject Slot chute into the solid waste bag.

1.1.3.3. The pipettor checks for successful ejection of the tip, and then proceeds to pick a new tip for the sample addition step, see Section 5.1.1.

1.1.4. Sample Addition

1.1.4.1. If the samples are in capped tubes, the pipettor will perform a piercing movement. The pipettor moves to a small distance from the top of the cap at a fast speed, then slows to 30% of the normal speed (for increased force at the z-motor) and pierces the cap a few millimeters deeper than necessary and then retracts to the normal final aspiration depth.

1.1.4.2. The pipettor then mixes the samples in the sample tube by aspirating and dispensing ~350ul of sample three separate times at a fixed height. The pressure traces are saved in the "ShearingPressure_..." file.

1.1.4.3. The pipettor then aspirates 400ul of sample, retracts to the opening of the sample tube, and aspirates a 15ul travelling air gap to reduce dripping. The pipettor routes the tip out from the sample bay to the sample dispense slot.

1.1.4.4. The pipettor then performs a "bubble pop" C-LLD. The height of the TCR liquid level is not regarded for this step. The intent is to pop any bubbles that may be above the true liquid level of the TCR. This C-LLD trace is stored in the "C-LLD1_..." file.

1.1.4.5. The pipettor retracts from the liquid level and starts another C-LLD. The height of the detected liquid level is converted into an LLD Value that is stored in the "LLDValues_..." file with PipettorDispense Sample PreLLD as the Command Description. The value stored is the detected height of TCR within the MTU tube, and it equals the number of z-motor steps from ZMax.

1.1.4.6. The pipettor dispenses the sample previously aspirated in step 5.1.4.3 and records the pressure trace to file "DispensePressureCurve_..." under Command Description "Sample."

1.1.4.7. The pipettor performs a C-LLD movement to pop any bubbles that may have occurred during the dispense. This trace is stored in the "C-LLD2_..." file.

1.1.4.8. After returning to a fixed height, one last C-LLD is performed in the MTU for sample. This C-LLD trace is stored in the "PostDispCapLLD_..." file and the correlated height is stored in "LLDValues_..." under Command Description "PipettorDispense Sample PostLLD." This value and the value from step 5.1.4.5 are compared. If the correlated volume difference of the PostLLD value and the PreLLD value falls outside of acceptable ranges for that sequence, then the test will be flagged as VVFS (Volume Verification Failure

Sample). The acceptable range is determined by sequence, for AC2 version 1.45.4, differences that fall outside of 200 to 600 µl of correlated volume are flagged.

1.1.5. Eject Tip / Pick New Tip

1.1.5.1. The tip used for sample aspiration and dispense is ejected at the Sample Eject Slot.

1.1.5.2. The Sample Arm repeats steps 5.1.1 through 5.1.5.1 for any test orders not in “Pending” state that remain on board.